A finite-field counterexample to the product-dimension conjecture

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Task 1: the published question. Conjecture 38 of Aliabadi [1], corresponding to Conjecture 5.1 of arXiv:2309.14664v3 and MathDB 360093, asserts the following implication. For finite-dimensional positive-dimensional K-subspaces A, B of a field extension L/K, assume

$$\dim_K(aM \cap A) + \dim_K(bM \cap B) \leq [M : K] + 1 \tag{1}$$

for all $a, b \in L$ and all nontrivial proper finite intermediate fields $K \subsetneq M \subsetneq L$. Then the proposed conclusion is

$$\dim_K\langle ST\rangle \geq \dim_K S + \dim_K T - 1 \qquad (S \subseteq A,\ T \subseteq B). \tag{2}$$

Here $\langle ST\rangle$ denotes the K-linear span of all products. The final published statement has no upper restriction on $\dim A + \dim B$ and explicitly excludes $M = L$ from (1).

Theorem 1. *Let*

$$K = \mathbb{F}_2, \quad L = \mathbb{F}_2[X]/(X^6 + X + 1), \quad \alpha = X \bmod (X^6 + X + 1),$$

and let

$$A = \langle \alpha, \alpha^2, \alpha^3, \alpha^4, \alpha^5\rangle_K, \qquad B = \langle \alpha, \alpha^2, \alpha^3\rangle_K.$$

These are proper subspaces of dimensions five and three, and neither contains 1*. They satisfy* (1) *for every scalar pair and every nontrivial proper intermediate field. Nevertheless,*

$$\dim_K\langle AB\rangle = 6 < 7 = \dim_K A + \dim_K B - 1.$$

Thus Conjecture 38 is false, including when its intermediate-field condition is nonvacuous and S, T are both nonzero.

Proof. Write $f = X^6 + X + 1$. A reducible degree-six polynomial has an irreducible factor of degree at most three. The complete list of monic irreducibles over $\mathbb{F}_2$ of these degrees is

$$X, \quad X + 1, \quad X^2 + X + 1, \quad X^3 + X + 1, \quad X^3 + X^2 + 1.$$

The respective remainders of f are

$$1, \quad 1, \quad X, \quad X^2 + X, \quad X^2 + 1.$$

They are all nonzero, so f is irreducible. Consequently L is a field with 64 elements, and $1, \alpha, \ldots, \alpha^5$ is a K-basis. This establishes the assertions about A, B and 1.

Every intermediate-field degree divides $[L : K] = 6$, by the tower law. The only possible nontrivial proper degrees are therefore two and three. The corresponding fields are the unique subfields

$$M_d = \{u \in L : u^{2^d} = u\} \qquad (d = 2, 3).$$

Indeed the Frobenius fixed sets are subfields; in $\mathbb{F}_{64}$ they have 2^d elements for $d \mid 6$. Alternatively, their elements can be checked directly using the encoding below:

$$M_2 = \{0, 1, 58, 59\}, \qquad M_3 = \{0, 1, 14, 15, 22, 23, 24, 25\}.$$

Each displayed set is closed under addition and multiplication and is the full root set of $U^{2^d} - U$. Any intermediate field of degree d consists of roots of that polynomial, which also proves uniqueness.

It remains to check every scalar translate, rather than just M_2, M_3 themselves. Encode $\sum_{j=0}^{5} c_j \alpha^j$ by the integer $\sum_{j=0}^{5} c_j 2^j$, where $c_j \in \{0, 1\}$. Integers in the next table are encodings of field elements, not residues modulo 64. Repeated multiplication with $\alpha^6 = \alpha + 1$ gives

$u \in B \setminus \{0\}$	2	4	6	8	10	12	14
u^3	8	3	59	24	54	17	15
u^7	6	20	26	59	28	31	1

For example, 2 encodes α, so its cube is encoded by 8, and its seventh power is $\alpha^2 + \alpha$, encoded by 6. Both power rows contain seven distinct elements.

If two nonzero elements u, v of B belonged to the same scalar translate bM_d, then $u/v \in M_d^*$ and hence $u^{2^d-1} = v^{2^d-1}$. The table forces $u = v$. Thus

$$\dim_K(bM_d \cap B) \le 1 \quad \text{for every } b \in L, \qquad d = 2, 3.$$

This also holds for $b = 0$. On the other hand, $\dim_K(aM_d \cap A) \le d$ for every $a \in L$. Adding these bounds proves (1) for all its quantified choices.

Finally, AB contains $\alpha^2, \alpha^3, \ldots, \alpha^7$. These six elements form a basis of L, since they are the image of $1, \alpha, \ldots, \alpha^5$ under the invertible multiplication map $u \mapsto \alpha^2 u$. Therefore $\langle AB \rangle = L$. Taking $S = A$ and $T = B$ proves the strict inequality in the theorem. □

Scope of the obstruction. The failure is the ambient-field bound. If the hypothesis also tested $M = L$ with $a = b = 1$, it would require $5 + 3 \le 6 + 1$, which this example violates. Replacing the conclusion by a bound capped at $[L : K]$, or adding a suitable dimension hypothesis, changes the question. No such modified assertion is resolved here. This is an obstruction in the published conjecture, not a difference introduced by MathDB. The source's related group proposition tests all nontrivial finite subgroups, including the ambient group when finite.

Verification and review. The accompanying standard-library Python verifier independently checks irreducibility, field arithmetic, both power rows, subfield closure, all $2 \cdot 64^2 = 8192$ scalar-pair instances of (1), and the product dimension. The finite certificate supplements the complete argument above. The review is by the author agent; no independent peer review or proof assistant verification is claimed. The dated literature audit records the sources checked and the limits of the search.

Task 2 (proposed). Determine the smallest extension degree over each finite base field that admits a counterexample with both A, B proper and at least one nontrivial proper intermediate field.

Task 3 (proposed). Classify the three-dimensional $\mathbb{F}_2$-subspaces of $\mathbb{F}_{64}$ meeting every scalar translate of $\mathbb{F}_4$ and $\mathbb{F}_8$ in dimension at most one, up to scalar multiplication and Frobenius automorphisms.

Task 4 (proposed). Determine whether (1) implies the capped inequality $\dim_K\langle ST\rangle \geq \min\{[L : K], \dim_K S + \dim_K T - 1\}$ for all nonzero S, T in arbitrary finite inseparable extensions.

Task 5 (proposed). Construct, or rule out, families satisfying (1) with proper A, B, nontrivial proper intermediate fields, and unbounded deficit $\dim_K A + \dim_K B - 1 - \dim_K\langle AB\rangle$.

References

[1] M. Aliabadi, *Conditions for matchability in groups and field extensions II*, Discussiones Mathematicae General Algebra and Applications **45** (2025), 135–157, doi:10.7151/dmgaa.1461. Conjecture 38, p. 154; preprint arXiv:2309.14664v3, Conjecture 5.1.